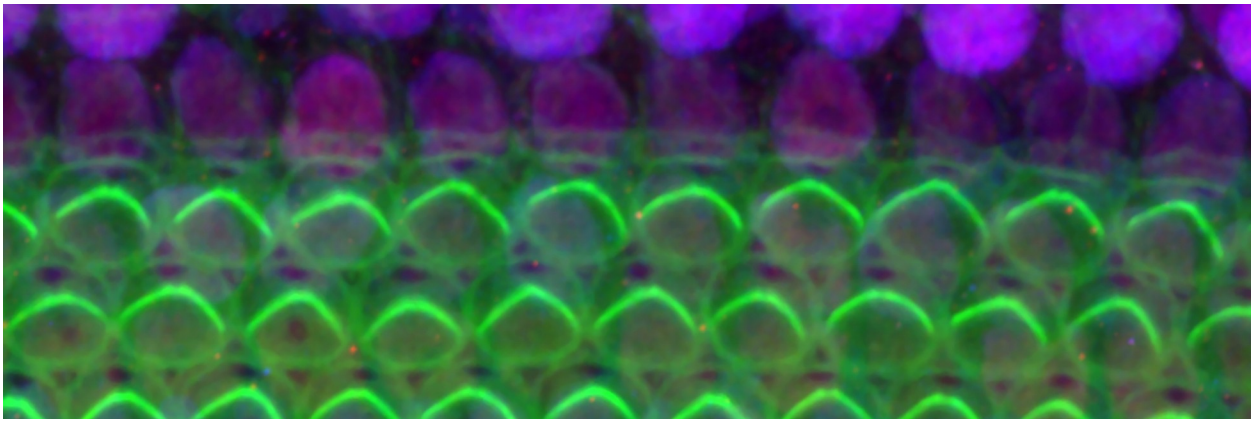


BISC 421: Neurobiology

Fall 2027 | *Units: _____*

Course Coordinator

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Office: Auditory Neuroscience Institute, Suite 500 (5th floor)
Office Hours: _____

Course Times & Locations

Component	Days & Times	Location
Lecture	Monday / Wednesday / Friday; 80 minutes Time: _____	_____
Wet Laboratory	Separate sections beginning Week 1 Days and times: _____	_____
Exams (4 total)	Separate session; time: _____ Planned: Sep 20, Oct 18, Nov 8 Exam 4: assigned appointment, Dec 8–15	_____

Course Description

Neurobiology asks how the cells and circuits of the nervous system give rise to sensation, movement, learning, and behavior. BISC 421 examines these questions through the biological mechanisms that allow neurons to generate electrical signals, communicate with one another, form connections during development, and change with experience. Topics range from ion channels and synapses to sensory processing, motor control, memory, brain states, and the cellular processes that contribute to neurological disease. Within each topic, we consider the mechanism, the experiments used to investigate it, and what the findings allow us to conclude.

The course is organized into four modules: cells and synaptic communication; development and plasticity; sensory processing and motor control; and neural systems, behavior, and dysfunction. These modules connect molecular and cellular processes with the operation of

neural circuits, drawing on foundational experiments and current research. Students learn to interpret physiological recordings and microscopy, evaluate competing explanations, and recognize the assumptions and limitations of the methods used to study the nervous system.

BISC 421 is co-taught, with Jordan Lopez serving as course coordinator. Lectures and separately scheduled wet laboratories bring together the biological concepts and the experimental work behind them. Laboratories begin in the first week and involve neural tissue, staining, microscopy, and physiological recordings. By the end of the semester, students should be able to explain how a neural mechanism works, identify the evidence supporting that explanation, and use experimental findings to develop a defensible scientific conclusion.

Prerequisites

Formal prerequisites: _____. Students should enter with a working background in cellular and molecular biology and basic chemistry. The course uses quantitative reasoning and introductory statistics to interpret experiments, while specialized neurobiological methods are developed through lecture and laboratory. Contact the course coordinator with questions about preparation or prerequisite eligibility.

Course Materials

Reference textbook: Kandel, E. R., Koester, J. D., Mack, S. H., and Siegelbaum, S. A., editors. Principles of Neural Science (6th ed.). McGraw Hill, 2021. This is a reference for the biological mechanisms discussed in class; no textbook purchase or paid courseware is required.

Assigned research articles and any designated textbook selections are required preparation. These readings provide the background for examining particular experiments, understanding the methods researchers used, and considering questions that remain unresolved. Lectures develop these ideas and extend beyond the reference textbook.

Lecture slides are the primary source of instruction, and exams are based on lecture content. Readings provide context and help you work through unfamiliar mechanisms and experimental findings, but the slides alone do not capture every explanation, example, or interpretation discussed in class.

This is a demanding upper-division course, and the material builds throughout the semester. Understanding neurobiology requires more than knowing the names of cells, structures, or signaling molecules; you need to explain how they contribute to a mechanism and how an experiment tests that explanation. Consistent preparation, careful work with figures, and regular review will be important across all four modules.

Required equipment: A laptop, a laboratory notebook, and access to the course site and the analysis tools specified for an assignment. Required laboratory clothing includes closed-toe shoes, a lab coat, and appropriate eye protection. Course equipment and biological materials are provided in the wet lab. Contact the coordinator or your TA if equipment or accessibility is a concern.

Course Objectives

- Explain how neurons, glia, ion channels, and synapses support nervous system function.

- Describe how neural circuits develop, become refined, and change with activity and experience.
- Connect cellular and circuit mechanisms with sensory processing, movement, memory, and behavior.
- Evaluate neurobiological experiments and interpret physiological recordings, microscopy, and quantitative evidence.
- Carry out wet-lab procedures responsibly and communicate conclusions supported by experimental observations.

Co-Instructors and Teaching Assistants

This is a co-taught course coordinated by Dr. Jordan Lopez, who oversees the course schedule, assessment policies, and final grades. Questions about material from a particular lecture should go to the instructor who taught it, while questions about laboratory logistics, reports, and assignment expectations should go to your assigned TA first. Concerns about scores or grading feedback, including requests for regrading, should be sent directly to the course coordinator at jlopez@usc.edu.

Role	Name	Email	Office Hours / Location
Co-Instructor	_____	_____	_____
Co-Instructor	_____	_____	_____
Wet-Lab TA	_____	_____	_____
Wet-Lab TA	_____	_____	_____
Exam Proctor	_____	_____	_____

Grading

This course has two components, lecture and laboratory. Each is scored out of 100 percent, and your final grade is 50 percent lecture and 50 percent laboratory. You must earn at least 60 percent in BOTH components to pass the course. A strong lecture score does not rescue a failing laboratory score, and the reverse is also true. The same grading policies apply across the co-taught modules.

Lecture Component (100%)

Assessment	Weight of Lecture Grade	Points
Exam 1 — Module 1 (Lectures 1–10)	20%	120
Exam 2 — Module 2 (Lectures 11–20)	20%	120
Exam 3 — Module 3 (Lectures 21–30)	20%	120
Exam 4 — Module 4 (Lectures 31–41)	20%	120
Lecture Project (choose one option)	20%	120
Total	100%	600

Laboratory Component (100%)

Assessment	Weight of Lab Grade
Lab Reports (8 counted of 10, 7.5% each)	60%
Individual Experimental Practical	10%
Methods Assignments (3 at 5% each)	15%
Wet-Lab Replication Project	15%
Total	100%

Dr. J's Grading Philosophy and Curve

Neurobiology requires you to connect biological mechanisms with experimental evidence. This is an upper-division course, and memorizing terminology alone is not sufficient; you must be able to explain how a cell or circuit works, evaluate the evidence supporting a proposed mechanism, and apply your understanding to unfamiliar results. The emphasis is on the quality of your reasoning and whether your explanation is consistent with the biology.

Exams and laboratory assignments reflect these expectations. Some assessments will be difficult, and raw percentages may be lower than you are accustomed to. Grades reflect demonstrated understanding, scientific accuracy, and the quality of your reasoning. Consistent effort and participation are important, but they do not substitute for meeting the course's academic standards.

If the final class average, calculated using the weighted lecture and laboratory grades before curving, is below 70%, a curve will be applied at the end of the course. If the class average is 70% or higher, no curve will be applied. The size of any adjustment will depend on overall class performance. A curve accounts for assessment difficulty while maintaining expectations for mastery; it does not guarantee a particular letter grade or a passing grade. Use early assessment results to identify gaps in your understanding, review feedback, and seek help promptly. Do not rely on a possible curve to compensate for missing work or consistently weak performance. You must still earn at least 60% in both the lecture and laboratory components to pass the course.

Grading Scale After Curving (changes year to year)

The percentage ranges below are approximate estimates, not fixed or guaranteed cutoffs. The grading scale may change from year to year, and final cutoffs for this semester may differ based on overall class performance and any curve applied under the policy above. Use this table as a general guide when estimating your grade.

Letter Grade	Approximate Percentage Range
A	90-100%
A-	85-89.9%
B+	80-84.9%
B	75-79.9%
B-	70-74.9%

Letter Grade	Approximate Percentage Range
C+	65-69.9%
C	60-64.9%
C-	55-59.9%
D	45-54.9%
F	Below 45%

Final grades are not rounded up. Letter grades will be assigned strictly according to the finalized cutoffs, which will be announced before final grades are assigned. Regardless of any adjustment to the scale, you must earn at least 60% in both the lecture and laboratory components to pass the course.

Exams

There are four exams, each worth 120 points. Exam 1 covers Lectures 1–10, Exam 2 covers Lectures 11–20, Exam 3 covers Lectures 21–30, and Exam 4 covers Lectures 31–41. Exams are not cumulative, although later topics build on earlier mechanisms. Planned dates are September 20, October 18, and November 8, with Exam 4 during the university examination period, December 8–15. Exams use a separate session so that regular lectures continue as scheduled. Exam time: _____. Room(s): _____. The final appointment and separate-session arrangements will be confirmed.

- **Number of questions:** Expect 65 questions totaling 120 points: 10 multiple-choice questions at 1 point each, 10 fill-in-the-blank questions at 1 point each, 35 short written-response questions at 2 points each, and 10 experimental or data-interpretation questions at 3 points each.
- **Exam versions:** Each exam comes in three versions (V1, V2, and V3). The three versions contain the same questions, with the order of questions and answer choices mixed within each lecture section.
- **Question order:** Exam questions are ordered by lecture. Questions on Lecture 1 come first, followed by questions on Lecture 2, and so on through the last lecture covered by that exam.
- **Room and version assignment:** Before each exam, a randomized algorithm assigns students to the reserved exam room or rooms and to one of three versions (V1, V2, or V3). Assignments are posted before the exam, and requests to change them will not be accepted. Approved accommodations are implemented through the university process.
- **Exam format:** In-person paper exams emphasizing written explanations of neural mechanisms, interpretation of experimental findings, and evaluation of experimental designs. Questions may ask you to explain a physiological response, interpret a micrograph or figure, compare evidence for competing mechanisms, or predict how a biological manipulation changes the system.

- **Expectations for answers:** Draw directly from lecture content, explain the relevant experiment and finding, and identify the evidence supporting your answer. Your explanation should connect the biological mechanism with the observed result rather than repeat terminology or slide text without showing understanding.
- **Missed exams:** Make-up exams are not given. If you miss one exam, a zero will initially be recorded. If the absence is approved for a valid, documented medical or official reason, that zero will be replaced at the end of the semester by the average of your other three exam scores. This replacement uses your other exam scores, not your final course grade. An undocumented or unapproved absence will remain a zero. Approved accommodations, religious observance, and university-required alternatives take precedence.
- Bring a photo ID and a pencil (YOU WILL BE VERIFIED). Phones, smartwatches, and notes are not allowed
- **Exams are closed-note** unless the class negotiates an alternative at the beginning of the semester (and/or before Exam I). The class may negotiate permission to bring **one 8.5 × 11-inch handwritten reference sheet (double sided, OK). In exchange, the class agrees to replace the multiple-choice questions with additional fill in the blank questions (any change needs to be negotiated with the class as a whole)**. Reference sheets must be handwritten; typed or printed material is not permitted.

New lecture content is introduced on exam days because exams use a separate session. Regular MWF lectures meet normally, and material introduced after an exam's coverage block is assessed on the next relevant exam.

Lecture Project (20% of Lecture Grade)

Each student completes one individual project on a neurobiological question covered in class. Choose ONE of the two options below and identify your choice in the proposal. Both options are worth 120 points and ask you to develop an explanation supported by experimental evidence.

Option A: Neural Data Analysis or Simulation Project

Investigate a neural mechanism using an approved dataset or a computational model grounded in experiments. Your project should connect the biological question with the measurements or model predictions and explain what the results tell us about the mechanism. The final work includes a five-page report, the analysis file or runnable model, and a five-minute recorded explanation, with original research providing the context for your interpretation.

Option B: Neurobiology Research Paper

Write a ten-page paper in APA 7th edition format on a focused neurobiological question, excluding the title page, abstract, figures, and references from the page count. The paper should bring together at least eight primary research articles, including recent work, to explain

a mechanism and critically evaluate the evidence supporting it. A series of unrelated article summaries is not sufficient; the paper needs to develop an argument about what the evidence supports and what remains uncertain. Include a five-minute recorded explanation of your conclusions.

Project Milestones

Milestone	Due
Proposal: chosen option, biological question, and initial research sources	Fri Sep 24, 11:59 PM
Progress check: preliminary analysis/model or argument and evidence	Fri Oct 22, 11:59 PM
Final report or paper, supporting files, and recorded explanation	Fri Nov 19, 11:59 PM

Laboratory Component

The laboratory is a required, graded component that begins in the first week and meets separately from lecture. These are wet laboratories in which you work with biological specimens and experimental preparations, carry out tissue staining and microscopy, and record physiological responses. Data analysis supports the bench work by connecting the observations with the mechanisms studied in lecture. Safety training and preparation are required before working with biological materials or equipment.

Laboratory safety and biological materials: Work only with the specimens, preparations, and equipment assigned for the session, and follow the TA's instructions for protective equipment, handling, and disposal. Biological materials and living preparations must be obtained and used through the university's applicable instructional and animal-care procedures. Course activities do not authorize independent collection, outside recruitment, or unsupervised experiments.

- **Lab Reports (10 labs, lowest 2 dropped, 7.5% each for the 8 counted).** Reports explain the biological question, how the experiment addressed it, what you observed, and how the findings relate to the mechanism under study. Each report is 4–6 pages, with appropriate figures, references, and an analysis file where relevant. The two dropped reports do not replace required safety training or completion of essential laboratory skills.
- **Individual Experimental Practical (10%).** During Week 11, you will demonstrate your understanding of one familiar wet-lab experiment from the five topics below. The practical brings together operation of the apparatus or preparation of the specimen with observation and explanation of the result. Preparation materials will be available by October 18, and your choice is due October 22, so you will know which procedures and concepts to prepare.
    - **Neuroanatomy and tissue organization:** Identify structures in a prepared neural specimen and explain their cellular organization.
    - **Nissl staining and neuronal morphology:** Prepare or examine a stained section and relate its cellular features to the tissue being studied.
    - **Immunofluorescence of neurons and glia:** Work with a familiar labeling preparation and explain what the signal and controls allow you to conclude.

    - **Compound action potentials:** Obtain and interpret an extracellular recording from an approved invertebrate preparation.
    - **Neuromuscular synaptic transmission:** Demonstrate a familiar recording procedure and explain the physiological response.
- **Methods Assignments (3 assignments, 5% each, 15% total).** These assignments develop your ability to quantify observations, interpret physiological recordings, and evaluate the controls used in microscopy and tissue-based experiments. They draw on the wet-lab work and provide practice explaining how the analysis supports a biological conclusion.
- **Wet-Lab Replication Project (15%).** Working in teams of 2–3, you will repeat an approved experiment introduced in the laboratory and examine whether the earlier result is reproduced. The project involves actual bench work with an approved tissue preparation or physiological recording, rather than a data-only exercise. Each student writes an individual four-page report, while teams demonstrate their experimental work and present the findings during the Week 15 laboratory sessions.

Lab Schedule and Due Dates

Lab / Session	Week	Topic	Due / Notes
Wet Lab 1	Week 1	Neuroanatomy and dissection of preserved neural specimens	Report due Sun Sep 5
Wet Lab 2	Week 2	Nissl staining and neuronal morphology	Report due Sun Sep 12
Wet Lab 3	Week 3	Immunofluorescence of neurons and glia	Report due Sun Sep 19
Wet Lab 4	Week 4	Extracellular recordings and compound action potentials	Report due Sun Sep 26
Wet Lab 5	Week 5	Ionic conditions and membrane excitability	Report due Sun Oct 3
Wet Lab 6	Week 6	Neuromuscular synaptic transmission	Report due Sun Oct 17; extended around Fall Recess
Wet Lab 7	Week 8	Neurite outgrowth in neuronal cell preparations	Report due Sun Oct 24
Wet Lab 8	Week 9	Retinal tissue and photoreceptor organization	Report due Sun Oct 31
Wet Lab 9	Week 10	Cochlear hair cells and sensory epithelia	Report due Sun Nov 7
Wet Lab 10	Week 12	Short-term synaptic plasticity	Report due Sun Nov 21; holiday-affected sections use an alternate wet-lab session
Wet-lab practical	Week 11	Individual practical from an earlier wet experiment	In assigned section Nov 1–5; project plan due Nov 5
Replication experiment	Week 13	Wet-lab replication: bench work and data collection	Data checkpoint due Fri Nov 19

Lab / Session	Week	Topic	Due / Notes
Bench demonstrations	Week 15	Replication verification and presentation at the bench	Individual report and team work due Fri Dec 3
Methods Assignment 1	Fri Sep 10	Quantifying cells and tissue organization	Due 11:59 PM
Methods Assignment 2	Fri Oct 1	Interpreting physiological recordings	Due 11:59 PM
Methods Assignment 3	Fri Oct 29	Microscopy controls and quantitative comparisons	Due 11:59 PM
Practical preparation	Mon Oct 18	Materials available for all five wet-lab options	At least two weeks before the practical
Practical topic choice	Fri Oct 22	Choose a familiar wet-lab experiment	Due 11:59 PM

Lab sections meet separately from lecture at the days, times, and locations listed under Course Times & Locations. All assignment deadlines are 11:59 PM Pacific time. No regular lab meets during Week 7 or Week 14. Sections affected by Veterans Day complete the required wet experiment in an arranged alternate session outside the university closure.

Lecture Recordings and Study Materials

Lectures are recorded. Recordings are released **one week before the exam that covers them**. I may release them two weeks before an exam at my discretion, but early release is not guaranteed and should not be expected. You are responsible for attending lecture, taking thorough notes, and studying throughout the semester. **Do not wait for recordings to begin preparing for an exam.**

Recordings from previous offerings may also be posted as supplemental study materials. This course changes between semesters, and older recordings may omit current topics or include material that is no longer covered. **This semester's lectures determine what is examinable.** You are responsible for identifying those differences using the current lectures and slides.

Recordings do not replace attendance or excuse arriving unprepared. Lecture content is dense and requires consistent engagement. Choosing to rely on recordings does not reduce the material you are expected to know or change the standards by which your work will be graded.

Study Expectations

This course covers substantial material on a fast schedule. Plan at least two hours of outside study for every hour of class, review each lecture within 48 hours, and work through experimental figures until you can explain the result without relying on the caption. Prepare for lab before arriving and review your observations while the experiment is still fresh. Study groups can help you test your understanding, but each student needs to explain the mechanisms and evidence independently.

Lecture & Lab Attendance

Attendance at lectures and wet laboratories is expected, but no separate attendance or participation grade is assigned. The wet-lab procedures and practical skills require in-person

work; if you miss a session, contact your TA promptly to arrange an approved way to complete the missed bench work. Contact the course coordinator directly about extensions, exceptional circumstances, and grading concerns.

Late Work

- Lab reports, methods assignments, and the Wet-Lab Replication Project: 5% deduction from the available points per calendar day late for the first 7 days. Work more than one week late is not accepted without an approved extension.
- **Experimental practical:** This assessment takes place in person during your assigned Week 11 laboratory session and is not eligible for the ordinary late-submission penalty. If an emergency or approved accommodation affects your attendance, contact the course coordinator as soon as possible to arrange an approved alternative. Missing the practical without an approved arrangement results in a zero.
- **Lecture project:** 30% deduction for any submission after 11:59 PM on the due date. Projects submitted more than 72 hours late are not accepted.

Extensions require instructor approval. Contact me before the deadline when possible, or as soon as reasonably possible in an emergency. Approved accommodations, religious observance, and university-required arrangements take precedence over ordinary late penalties.

Academic Integrity and Use of AI Tools

Follow the university's standards of academic integrity. Cheating, plagiarism, fabricated measurements or citations, unauthorized collaboration, and unauthorized exam materials are handled under university procedures. Generative AI may not write report prose, the research paper, individual scientific interpretations, or exam answers. Coding or software-troubleshooting assistance is permitted only when authorized and disclosed; verify the output and retain the original files. Shared bench work is permitted for specified team activities, while individual reports and interpretations must reflect your own understanding. Do not upload classmates' information or restricted course data to external AI services.

Accommodations

Students seeking accommodations should contact the Office of Student Accessibility Services as early as possible and provide the approved notice through the university process. Coordinate lecture, wet-lab, equipment, deadline, and exam arrangements with the course coordinator so that the course team can plan appropriately. A course deadline does not cancel an approved accommodation. OSAS: https://osas.usc.edu/.

Communication

Questions about biological mechanisms, experimental findings, and laboratory observations are expected. Contact the instructor responsible for a lecture about its content, and your assigned TA about routine laboratory logistics, reports, and analysis. Contact the course coordinator, Jordan Lopez, at jlopez@usc.edu for grading concerns and regrade requests, unresolved questions, accommodations, or confidential issues.

I will make every effort to respond to emails and messages promptly; however, because I teach multiple courses and run an active research laboratory, response times may be delayed. Students should generally expect a response within one business day. Messages sent in the evening, overnight, or on weekends are typically addressed the following business day. Use clear subject lines that include the course number and topic, such as “BISC 421: Lab 3 report question,” and include the context needed to understand the issue. For questions about a co-instructor’s lecture, identify the lecture number and contact that instructor directly.

Absences and Communication

I understand that students balance academic responsibilities with work, family obligations, and other commitments. If you anticipate an absence or experience an emergency, contact the course coordinator as soon as possible. For laboratory absences, notify your TA promptly so that an appropriate arrangement can be made for the required wet-lab work. Communicating early helps us address the situation before missed work accumulates.

There is no separate lecture or laboratory attendance grade, but you remain responsible for missed instruction and graded work. Recordings support review and do not replace laboratory procedures, the experimental practical, or the bench work required for the replication project. Missed assignments and assessments remain subject to their applicable policies.

Course Calendar

Lectures meet Monday, Wednesday, and Friday for 80 minutes, beginning August 23 and ending December 3. Every instructional meeting introduces a new topic across four modules. Exams 1–3 are planned for September 20, October 18, and November 8 in a separate session, with the final exam during December 8–15. Session times and rooms remain unassigned. Assignment deadlines are 11:59 PM Pacific time unless otherwise stated. University closures follow USC’s Fall 2027 academic calendar: https://www.usc.edu/academic-calendar/.

Date (2027)	Session	Topic	Assessment / Notes
Mon Aug 23	Wet Lab 1 · Wk 1	Neuroanatomy and dissection of preserved neural specimens	Meets Aug 23–27
Mon Aug 23	Lecture 1	Neurons, cell types, and the organization of the nervous system	Module 1 begins
Wed Aug 25	Lecture 2	Glia, myelination, and neuronal support	
Fri Aug 27	Lecture 3	Ion gradients and the resting membrane potential	
Mon Aug 30	Wet Lab 2 · Wk 2	Nissl staining and neuronal morphology	Meets Aug 30–Sep 3
Mon Aug 30	Lecture 4	Ion channels: selectivity, gating, and channelopathies	
Wed Sep 1	Lecture 5	Action potentials and axonal conduction	
Fri Sep 3	Lecture 6	Dendrites and the integration of synaptic inputs	
Sun Sep 5	Lab 1 report due	Neuroanatomy and dissection of preserved neural specimens	11:59 PM
Mon Sep 6	Holiday	Labor Day	No lecture or lab
Tue Sep 7	Wet Lab 3 · Wk 3	Immunofluorescence of neurons and glia	Meets Sep 7–10
Wed Sep 8	Lecture 7	Synaptic vesicles and the mechanisms of transmitter release	

Date (2027)	Session	Topic	Assessment / Notes
Fri Sep 10	Lecture 8	Excitatory and inhibitory synaptic transmission	
Fri Sep 10	Methods Assignment 1 due	Quantifying cells and tissue organization	11:59 PM
Sun Sep 12	Lab 2 report due	Nissl staining and neuronal morphology	11:59 PM
Mon Sep 13	Wet Lab 4 · Wk 4	Extracellular recordings and compound action potentials	Meets Sep 13–17
Mon Sep 13	Lecture 9	Neuromodulators, G-protein-coupled receptors, and intracellular signaling	
Wed Sep 15	Lecture 10	Measuring and manipulating neurons: electrophysiology, imaging, and optogenetics	
Fri Sep 17	Lecture 11	Neural induction and the patterning of the nervous system	Module 2 begins
Sun Sep 19	Lab 3 report due	Immunofluorescence of neurons and glia	11:59 PM
Mon Sep 20	Wet Lab 5 · Wk 5	Ionic conditions and membrane excitability	Meets Sep 20–24
Mon Sep 20	Lecture 12	Neural stem cells, neurogenesis, and cell-fate specification	
Mon Sep 20	EXAM 1 (planned)	Lectures 1–10	Separate session; time/room TBD
Wed Sep 22	Lecture 13	Neuronal migration and cortical development	
Fri Sep 24	Lecture 14	Axon guidance and target recognition	
Fri Sep 24	Project proposal due	Lecture project question, option, and sources	11:59 PM
Sun Sep 26	Lab 4 report due	Extracellular recordings and compound action potentials	11:59 PM
Mon Sep 27	Wet Lab 6 · Wk 6	Neuromuscular synaptic transmission	Meets Sep 27–Oct 1
Mon Sep 27	Lecture 15	Synapse formation and the assembly of neural circuits	
Wed Sep 29	Lecture 16	Activity-dependent circuit refinement and critical periods	
Fri Oct 1	Lecture 17	Long-term potentiation and long-term depression	
Fri Oct 1	Methods Assignment 2 due	Interpreting physiological recordings	11:59 PM
Sun Oct 3	Lab 5 report due	Ionic conditions and membrane excitability	11:59 PM
Mon Oct 4	Week 7 laboratory	Fall Recess week	No regular lab
Mon Oct 4	Lecture 18	Homeostatic plasticity and excitation–inhibition balance	
Wed Oct 6	Lecture 19	Microglia, immune signaling, and synaptic pruning	
Thu Oct 7	Holiday	Fall Recess: Oct 7–8	No classes during closure
Mon Oct 11	Wet Lab 7 · Wk 8	Neurite outgrowth in neuronal cell preparations	Meets Oct 11–15

Date (2027)	Session	Topic	Assessment / Notes
Mon Oct 11	Lecture 20	Axon regeneration and neural repair	
Wed Oct 13	Lecture 21	Sensory transduction and neural population coding	Module 3 begins
Fri Oct 15	Lecture 22	Touch, mechanotransduction, and proprioception	
Sun Oct 17	Lab 6 report due	Neuromuscular synaptic transmission	11:59 PM
Mon Oct 18	Wet Lab 8 · Wk 9	Retinal tissue and photoreceptor organization	Meets Oct 18–22
Mon Oct 18	Lecture 23	Nociception and the neural control of pain	
Mon Oct 18	EXAM 2 (planned)	Lectures 11–20	Separate session; time/room TBD
Mon Oct 18	Practical preparation	Materials available for all five options	
Wed Oct 20	Lecture 24	The retina: phototransduction and early visual processing	
Fri Oct 22	Lecture 25	Thalamic and cortical circuits for vision	
Fri Oct 22	Project progress due	Preliminary work and evidence	11:59 PM
Fri Oct 22	Practical choice due	Choose a familiar wet experiment	11:59 PM
Sun Oct 24	Lab 7 report due	Neurite outgrowth in neuronal cell preparations	11:59 PM
Mon Oct 25	Wet Lab 9 · Wk 10	Cochlear hair cells and sensory epithelia	Meets Oct 25–29
Mon Oct 25	Lecture 26	Hearing: cochlear transduction and auditory circuits	
Wed Oct 27	Lecture 27	Olfaction and taste: the biology of chemical sensing	
Fri Oct 29	Lecture 28	Spinal motor circuits and the generation of movement	
Fri Oct 29	Methods Assignment 3 due	Microscopy controls and quantitative comparisons	11:59 PM
Sun Oct 31	Lab 8 report due	Retinal tissue and photoreceptor organization	11:59 PM
Mon Nov 1	Week 11 laboratory	Wet-lab practical and replication planning	In assigned section Nov 1–5
Mon Nov 1	Lecture 29	Basal ganglia circuits and action selection	
Wed Nov 3	Lecture 30	The cerebellum: coordination, balance, and motor learning	
Fri Nov 5	Lecture 31	Hypothalamic and autonomic circuits for homeostasis	Module 4 begins
Fri Nov 5	Replication plan due	Approved experiment and prediction	11:59 PM
Sun Nov 7	Lab 9 report due	Cochlear hair cells and sensory epithelia	11:59 PM
Mon Nov 8	Wet Lab 10 · Wk 12	Short-term synaptic plasticity	Meets Nov 8–12
Mon Nov 8	Lecture 32	Circadian clocks, sleep, and arousal	

Date (2027)	Session	Topic	Assessment / Notes
Mon Nov 8	EXAM 3 (planned)	Lectures 21–30	Separate session; time/room TBD
Wed Nov 10	Lecture 33	Stress biology and the hypothalamic–pituitary–adrenal axis	
Thu Nov 11	Holiday	Veterans Day (Thursday)	No lab during closure; MWF lectures unchanged
Fri Nov 12	Lecture 34	Dopamine, reward prediction, and reinforcement learning	
Mon Nov 15	Week 13 laboratory	Wet-lab replication experiments	Data checkpoint due Nov 19
Mon Nov 15	Lecture 35	Hippocampal circuits for spatial and episodic memory	
Wed Nov 17	Lecture 36	Memory engrams, consolidation, and retrieval	
Fri Nov 19	Lecture 37	Amygdala circuits for threat and safety learning	
Fri Nov 19	Lecture project final due	Report/paper, files, and recorded explanation	11:59 PM
Fri Nov 19	Replication data check	Bench measurements and preliminary findings	11:59 PM
Sun Nov 21	Lab 10 report due	Short-term synaptic plasticity	11:59 PM
Mon Nov 22	Week 14 laboratory	Thanksgiving week	No regular lab
Mon Nov 22	Lecture 38	Prefrontal circuits for working memory and cognitive control	
Wed Nov 24	Holiday	Thanksgiving: Nov 24–28	No lecture or lab
Mon Nov 29	Week 15 laboratory	Bench verification and replication demonstrations	Final work due Dec 3
Mon Nov 29	Lecture 39	Neuropeptides and the neural regulation of social behavior	
Wed Dec 1	Lecture 40	Epilepsy: pathological excitability and network synchrony	
Fri Dec 3	Lecture 41	Neurodegeneration: protein homeostasis and selective neuronal vulnerability	
Fri Dec 3	Replication final due	Individual four-page report and team work	11:59 PM
Sat Dec 4	Study days	December 4–7	No regular lectures or labs
Wed Dec 8	EXAM 4 window	Lectures 31–41; university exams Dec 8–15	Assigned appointment TBD

Readings Calendar

Specific textbook selections and research articles will be added to the readings calendar separately. Assigned selections are required preparation and support the mechanisms and experiments discussed in class. The current lectures determine exam coverage; the reference textbook provides additional biological context.

Final Word

This course asks you to explain the nervous system through its biology and the experiments that reveal how it works. By connecting cells, circuits, development, and behavior with observations from the wet laboratory, you will develop a foundation for reading neurobiological research and evaluating its conclusions. Come prepared to ask what an experiment tested, what its results show, and what would be needed to distinguish the remaining explanations.

Banner: Newborn mouse cochlear hair cells and supporting cells, by Elizabeth Carroll Driver, Ph.D., National Institute on Deafness and Other Communication Disorders, NIH; public domain. Cropped banner. Source: https://commons.wikimedia.org/wiki/File:Mouse_Cochlea_(42112495551).jpg